

Yelp Dataset Sentiment Analysis

Phase 2: Project Plan

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Literature Review Recap – Motivation

- Yelp hosts millions of restaurant reviews.
- Star ratings summarize satisfaction but are **subjective and inconsistent**.
- Automatically predicting ratings from text helps:
 - Users gauge sentiment quickly
 - Businesses identify improvement areas
 - Researchers study drivers of satisfaction

Restaurant: XYZ, Kitchener, N2G4Z6, Canada

Rating: ● ● ● ○ ○

I've been to XYZ a bunch of times. It's a decent place. Nice food, lots of variety! The place is really small though, so you almost never find a spot to sit and eat. The service is also slow at times.

Source: Asghar et al.

Literature Review Recap — Findings

- What prior work found:
 - Binary sentiment classification (positive vs. negative) → High accuracy ($\approx 90\%$)
 - Multi-class star prediction (1–5) → Lower accuracy ($\approx 60\text{--}65\%$)
 - Best features: **TF-IDF + n-grams**
 - Best models: **Logistic Regression, SVM**
- Deep models (CNNs, Bi-LSTMs, BERT) show promise but struggle with:
 - **Middle ratings (2–3 stars)**
 - **Imbalanced data** (too many 4–5s)
 - **Sarcasm and slang**

Literature Review Recap — Key Gaps

- Context understanding still limited (“Great job burning my pizza!”).
- 4–5 star dominance causes skewed learning.
- Ordinal sentiment analysis underexplored.
- Deep models improve nuance but need more explainability.



Source: Asghar et al.



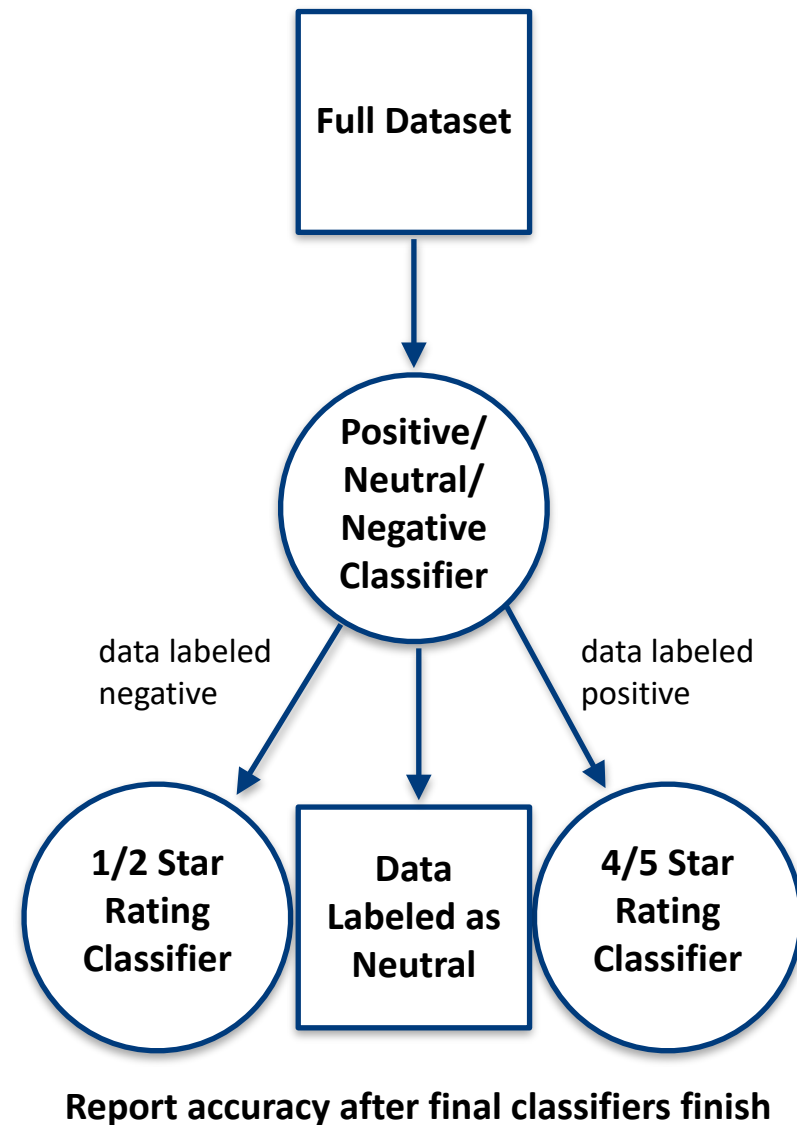
Source: Fan & Khademi

Project Goal

- **Primary Goal:** Predict restaurant review star ratings (1–5) from text.
- **Secondary Goals:**
 - Compare **text feature extraction methods**: n-grams, TF-IDF, embeddings.
 - Evaluate **model families**: linear, tree-based, neural, ensemble.
 - Handle **class imbalance** and analyze **per-class performance**.
 - Explore **regression** and **ordinal approaches** to better model star order.

Algorithms & Methods

- **Baseline Models:**
 - Logistic Regression, Naive Bayes, SVM, Random Forest
- **Advanced Models:**
 - Neural networks (MLP + BERT embeddings)
 - Ensemble / stacking combinations
- **Experimental Variations:**
 - Regression vs. Classification
 - Ordinal Logistic Regression
 - *** **Multi-stage models (e.g., sentiment first → star regression)** ***



Data Preparation

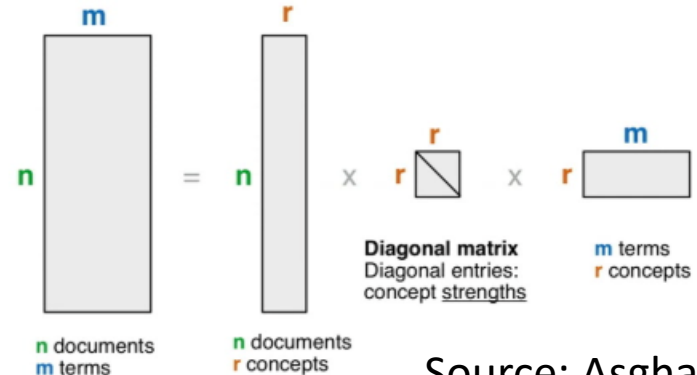
- **Dataset:** Yelp Open Dataset (subset: specific state restaurant reviews).
- **Tasks:**
 - Filter restaurant reviews.
 - Clean and normalize text (lowercasing, punctuation removal, stopwords removal).
 - Derive auxiliary features (e.g., review length, punctuation counts).
 - Split data into training, validation, and test sets. (90% training, 10% test)
- **Output:** Cleaned, balanced dataset ready for modeling.



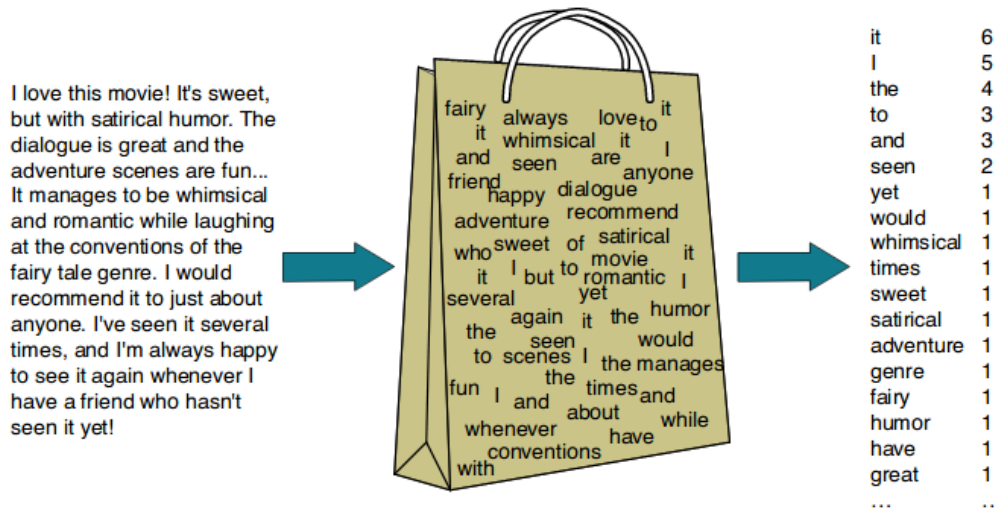
Source: Columbia University

Feature Engineering

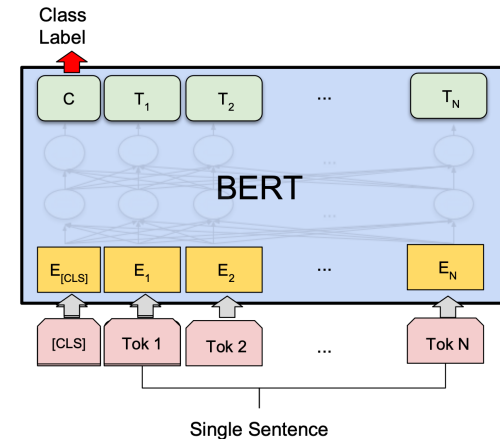
- May experiment with multiple text representations:
 - **TF-IDF** (uni-, bi-, tri-grams)
 - **Latent Semantic Indexing**
 - Contextual embeddings (**BERT**)



Source: Asghar et al.



Source: Medium



Source: Wandb

Experimental Plan: Week 1 (11/3 - 11/9)

- **Tasks**

- Implement and train baseline models: Logistic Regression, Naive Bayes, SVM, Random Forest, using n-grams with TF-IDF (both binary and 1-5 RRP).
- Evaluate and document baseline performance (accuracy, F1).
- Set up pipeline for layered (sentiment) classification (positive/negative/neutral segmentation).

- **Deliverables**

- Baseline metrics

Experimental Plan: Week 2 (11/10 - 11/16)

- **Tasks**

- Implement layered classifier structure: first-stage multiclass (pos/neg/neutral), then separate binary classifiers for positives (4 vs 5) and negatives (1 vs 2).
- Integrate contextual embeddings (BERT-based features).

- **Deliverables**

- Layered classifier pipeline functional.
- BERT embeddings incorporated.

Experimental Plan: Week 3 (11/17 - 11/23)

- **Tasks**

- Implement ensemble models: weighted voting and stacking across best baseline and advanced classifiers.
- Test ordinal and regression-based models (Ordered Logistic Regression, RandomForestRegressor, SVMRegressor).
- Analyze per-class recall and confusion matrices, especially for mid-range ratings.
- Start hyperparameter tuning for all models.

- **Deliverables**

- Results from ensemble and ordinal models.
- Comparative metrics and updated confusion matrices.
- Record of hyperparameter search and preliminary best models.

Experimental Plan: Week 4 (11/24 - 11/30)

- **Tasks**

- Finalize all models and run full evaluation against test set.
- Consolidate all results; analyze failure cases and limitations (e.g., where models confuse 2 vs 3 stars).
- Complete project report and prep presentation.

- **Deliverables**

- Final metrics table and visualizations (e.g., confusion matrix).
- Short analysis of key insights, limitations, and suggested future work.
- Finished full project report and slides.

Baseline Results (subsample n=500,000)

Features = Unigram + Bigram + TF_IDF w/ max_features = 10,000

Model	Classification Type	Accuracy (%)	Macro-F1
Logistic Regression	1-5 RRP	67.66	0.5969
Naive Bayes	1-5 RRP	64.15	0.5187
SVM	1-5 RRP	44.70	0.4147
Random Forest	1-5 RRP	51.96	0.2869
Ordinal Logistic Regression	1-5 RRP	60.92	0.5268
Logistic Regression	Pos. / Neu. / Neg.	87.43	0.7398
Naive Bayes	Pos. / Neu. / Neg.	84.45	0.6494
SVM	Pos. / Neu. / Neg.	60.63	0.5081
Random Forest	Pos. / Neu. / Neg.	75.83	0.4622
Ordinal Logistic Regression	Pos. / Neu. / Neg.	85.66	0.7133

Sources

Sentiment Analysis on Product Reviews Using Machine Learning Techniques; Rajkumar S. Jagdale, Vishal S. Shirsat and Sachin N. Deshmuk

Yelp Dataset Challenge: Review Rating Prediction; Nabiha Asghar

Sentiment Analysis: A Systematic Case Study with Yelp Scores; Wenping Wang Et al.

Ensemble Sentiment Analysis Using Bi-LSTM and CNN; Puneet Singh Lamba Et al.

Sentiment Analysis of Restaurant Reviews using Combined CNN-LSTM; Naimul Hossain Et al.

Sentiment Analysis of Yelp Reviews by Machine Learning; Hemalatha S, Ramathmika

Sentiment Analysis on Food Review using Machine Learning Approach; Nourin Islam, Ms. Nasrin Akter, Abdus Sattar

Sentiment Analysis using Machine Learning Techniques on Python; Ratheee Et al.

Sentiment Analysis of Yelp's Ratings Based on Text Reviews; Xu Et al.

Predicting a Business' Star in Yelp from Its Reviews' Text Alone; Fan, Khademi